

Route G-R3: A Density Candidate and Two Relational Clocks

An explicit interaction, not a fitted slowdown law

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Abstract

We define an excitation-energy density relative to a declared counting measure, couple it to one clock, and compute its readings relative to another in an exact finite stationary state. The resulting relative rate is observable under fixed calibration, but its sign is determined by the interaction, not by the word density. A separate dispersive exchange calculation admits both signs. Unobserved source fluctuations produce a distribution of conditional phases, not generally a single rate evaluated at the mean density. This closes one bounded candidate test, not Chen's density-time claim, emergent gravity, or a universal time-dilation principle.

1 Define the object before assigning it dynamics

Retain the G-R1 warning: energy statistics alone do not determine a relational history. A candidate full information object would include the joint state, event algebra, recording instruments and reference measure. We do not identify any one scalar with all of that information.

Take an addressed matter subsystem D with ground state $|0\rangle$, excited state $|1\rangle$, and positive excitation energy ϵ_D . Fix the measure of this one operational cell to $\nu(D) = 1$. Define

$$N_D = |1\rangle\langle 1|, \quad H_D - E_{D,g}\mathbb{I} = \epsilon_D N_D, \quad \hat{E}_d = \frac{\epsilon_D N_D}{\nu(D)}, \quad E_0 = \frac{\epsilon_D}{\nu(D)}. \quad (1)$$

Units are energy per declared cell, *not* energy per spatial volume. For several declared cells, an additive operator measure can be $\mu_E(S) = \sum_{j \in S} \epsilon_j N_j$ with counting measure $\nu(S) = |S|$; its average is $\mu_E(S)/\nu(S)$. The addressed set, weights and coupling must be fixed together, not changed to fit a rate. This one-cell test cannot empirically distinguish density from total excitation energy. Ground subtraction prevents an arbitrary source energy zero from changing the interaction.

Let B be a three-level clock, $H_B = \epsilon N_B$ with $N_B = \text{diag}(0, 1, 2)$. Choose the single interaction

$$V = \kappa H_B \frac{\hat{E}_d}{E_0} = \kappa \epsilon N_B N_D, \quad H_{BD} = \epsilon N_B (1 + \kappa N_D) + \epsilon_D N_D. \quad (2)$$

Tensor identities are suppressed. We impose $\kappa > -1$ for positive clock gaps. $[N_D, H_{BD}] = 0$: this is a dispersive, excitation-preserving coupling. κ is a declared coupling, not a fitted time function. A reference clock A is shielded from D . This asymmetry is an input.

2 One common constraint and an exact finite solution

We explicitly extend the G-R1 stationary model to $A \otimes B \otimes D$. We do not equate it with G-R2's recording-history constraint. For an exactly reproducible card fix

$$\epsilon_D = \epsilon, \quad \delta = \epsilon/4, \quad \kappa = 1/4, \quad D_A = 15, \quad E_* = 14\delta, \quad H_A = \sum_{j=0}^{14} (14-j)\delta |j\rangle\langle j|_A. \quad (3)$$

The alternative audit cards $\kappa = 0, -1/4$ use the *same* A spectrum, calibration and readout. These rational choices make the finite recurrence exact; they are not measurements or fits. For $n = 0, 1, 2$ and $d = 0, 1$, define

$$e_{nd} = \epsilon n(1 + \kappa d) + \epsilon_D d, \quad j_{nd} = e_{nd}/\delta = 4n(1 + \kappa d) + 4d. \quad (4)$$

All j_{nd} are integers between 0 and 14 on these cards. For normalized a_n, c_d , put

$$K = H_A + H_{BD} - E_* \mathbb{I}, \quad (5)$$

$$|\Psi\rangle = \sum_{n,d} a_n c_d |j_{nd}\rangle_A |n\rangle_B |d\rangle_D, \quad K|\Psi\rangle = 0. \quad (6)$$

Each term has total energy E_* ; coincident j_{nd} cause no normalization problem because the BD labels remain orthogonal. Both local energies and the combined H_{BD} are nonnegative. The matched state is an assumed preparation, not a dynamically derived attractor. No arbitrary incommensurate-spectrum extension is claimed.

Use the reference phase POVM

$$|\phi\rangle_A = D_A^{-1/2} \sum_{j=0}^{14} e^{ij\phi} |j\rangle, \quad F_A(d\phi) = \frac{D_A}{2\pi} |\phi\rangle\langle\phi|_A d\phi. \quad (7)$$

Its integral is $\mathbb{I}_A$; phase density is uniformly $1/(2\pi)$. Normalized conditioning of (6) gives

$$|\psi_{BD}(\phi)\rangle = \sum_{n,d} a_n c_d e^{-ie_{nd}\phi/\delta} |n, d\rangle, \quad i\delta \partial_\phi |\psi_{BD}\rangle = H_{BD} |\psi_{BD}\rangle. \quad (8)$$

This comes directly from the constraint, without inserting an external physical time. Only after calibration write $\tau_A = \hbar\phi/\delta$. For a generic interaction involving the reference itself the simple conditional equation need not hold [1]. Here A is decoupled from BD except through the stationary correlation, so it holds exactly.

For completeness, if an event effect F fails to commute with K , the energy-sector pinching

$$\mathcal{P}_K(F) = \sum_\ell P_\ell F P_\ell, \quad K = \sum_\ell k_\ell P_\ell \quad (9)$$

is positive, preserves POVM completeness, commutes with K , and gives the same expectation in Ψ . The joint probabilities thus admit constraint-invariant effects. This is not a microscopic implementation of a nonlocal measurement apparatus.

3 Compare two readings, not two coordinate labels

Set $a_n = 1/\sqrt{3}$. Clock B has phase states and POVM

$$|\theta\rangle_B = \frac{1}{\sqrt{3}} \sum_{n=0}^2 e^{-in\theta} |n\rangle, \quad F_B(d\theta) = \frac{3}{2\pi} |\theta\rangle\langle\theta|_B d\theta. \quad (10)$$

The Fourier sign is opposite to that of A , whose Hamiltonian has the complementary spectrum. Neither phase is a geometric rotation. Conditional on source outcome d , (8) becomes, up to a source-only phase, $|\theta_d(\phi)\rangle_B$ with

$$\theta_d(\phi) = \frac{\epsilon}{\delta} (1 + \kappa d) \phi \pmod{2\pi}. \quad (11)$$

The actual likelihood, rather than a zero-width clock reading, is

$$p(\theta \mid \phi, d) = \frac{1}{6\pi} \left| \sum_{n=0}^2 e^{in[\theta - \theta_d(\phi)]} \right|^2 = \frac{[1 + 2 \cos(\theta - \theta_d)]^2}{6\pi}. \quad (12)$$

It integrates to one. The complete joint density is $p(\phi, \theta, d) = |c_d|^2 p(\theta \mid \phi, d) / (2\pi)$. On a fixed three-outcome grid $\theta_j = 2\pi j/3$, the corresponding projective probabilities are $[1 + 2 \cos(\theta_j - \theta_d)]^2/9$.

Calibrate the uncoupled B dial by $\tau_B = \hbar\theta/\epsilon$, keeping that calibration when D is excited. On a specified unwrapped phase branch, the shift of the likelihood peak gives

$$r_{B/A}(d) = \frac{d\tau_B}{d\tau_A} = \frac{\delta}{\epsilon} \frac{d\theta_d}{d\phi} = 1 + \kappa d = 1 + \kappa \frac{E_d(d)}{E_0}. \quad (13)$$

This is a result of (2), not an independent phenomenological slowdown fit. At $d = 1$ the three audit cards give $r = 3/4, 1, 5/4$: either direction is possible. Finite POVM width requires repeated calibrated comparisons. One event does not reveal an exact rate or a unique cycle count. Use a predeclared short branch, or the event/epoch distinction of G-R2, to avoid unwrapping ambiguity.

Two invariance checks matter. A common energy rescaling $(\epsilon, \epsilon_D, \delta, E_*) \mapsto s(\epsilon, \epsilon_D, \delta, E_*)$ rescales K but leaves Ψ , all phase probabilities and r unchanged. A passive relabeling of a dial, with its POVM and calibration transformed consistently, also cannot change these probabilities. Recalibrating B separately in every density sector would instead remove the comparison one set out to measure.

For the same commuting conditional-generator structure, if both clocks receive factors $1 + \kappa_A d$ and $1 + \kappa_B d$, then

$$r_{B/A} = \frac{1 + \kappa_B d}{1 + \kappa_A d}. \quad (14)$$

Equal factors cancel. This elementary common-mode diagnostic is not a claim to have solved an arbitrary interaction with the finite reference. Only a differential response is observable by the two-clock comparison.

4 Fluctuating density: why the mean is not one effective clock

Let $w_d = |c_d|^2$. When D is not read, the reduced clock state is

$$\rho_B(\phi) = \sum_d w_d |\theta_d(\phi)\rangle \langle \theta_d(\phi)|, \quad (15)$$

$$[\rho_B(\phi)]_{nm} = \frac{1}{3} e^{-i(n-m)x} \sum_d w_d e^{-i(n-m)\kappa dx}, \quad x = \epsilon\phi/\delta. \quad (16)$$

The last factor is a characteristic function of the density distribution. It is not generally $\exp[-i(n-m)\kappa\langle d\rangle x]$. For small κx ,

$$\log\langle e^{-iyd}\rangle = -iy\langle d\rangle - \frac{1}{2}y^2 \text{Var}(d) + O(y^3), \quad y = (n-m)\kappa x. \quad (17)$$

The mean shifts phase, while the variance first changes visibility. Replacing the operator by its mean drops precisely that information. This respects, rather than overturns, the G-R1 density-only warning.

For $w_0 = w_1 = 1/2$, the adjacent coherence is

$$[\rho_B]_{10} = \frac{1}{3} e^{-ix} \frac{1}{2} (1 + e^{-i\kappa x}). \quad (18)$$

With $\kappa = 1/4$ at $\phi = \pi$, $x = 4\pi$; adjacent coherences vanish, but $[\rho_B]_{20} = 1/3$ and $\text{tr} \rho_B^2 = 5/9$. This is neither a pure clock with an average rate nor a fully dephased state. A coherent pure source entangles with B ; a classical source mixture yields the same reduced B statistics. These data alone do not witness quantum source coherence.

An exact probability test uses $\phi = \pi/2$ on the same card. Then $\theta_0 = 2\pi$ and $\theta_1 = 5\pi/2$. In the fixed B grid:

source	p_0	p_1	p_2
$d = 0$	1	0	0
$d = 1$	1/9	$(1 + \sqrt{3})^2/9$	$(1 - \sqrt{3})^2/9$
unread equal weights	5/9	$(1 + \sqrt{3})^2/18$	$(1 - \sqrt{3})^2/18$

In contrast, replacing d by $1/2$ in the Hamiltonian predicts

$$p_0^{\text{mean substitution}} = \frac{(1 + \sqrt{2})^2}{9} = \frac{3 + 2\sqrt{2}}{9} \neq \frac{5}{9}. \quad (19)$$

This tests the approximation without fitting any data. The source projector measurement itself commutes with H_{BD} , although conditioning on its result selects a branch. A phase measurement on the clock is a separate instrument with backreaction; it cannot be silently replaced by nondisturbing, continuously available readings.

5 A microscopic sign check, separate from the exact rational card

Use the Jaynes–Cummings exchange model, in frequency units ($\hbar = 1$ in this section):

$$H_{\text{JC}} = \omega b^\dagger b + \Omega N_D + g(b\sigma_+ + b^\dagger\sigma_-), \quad \Delta = \Omega - \omega. \quad (20)$$

This is an established idealized model [2], not a hardware calibration. For $g\sqrt{n+1}/|\Delta| \ll 1$, put $X = b\sigma_+$ and $S = (g/\Delta)(X - X^\dagger)$. Direct algebra gives

$$[S, H_0] = -V, \quad [X, X^\dagger] = (2N_D - 1)n + N_D, \quad n = b^\dagger b. \quad (21)$$

Hence the second-order transformed Hamiltonian is

$$\begin{aligned} H_{\text{eff}} &= H_0 + \tfrac{1}{2}[S, V] + O(g^3/\Delta^2) \\ &= (\omega - \chi)n + (\Omega + \chi)N_D + 2\chi n N_D + O(g^3/\Delta^2), \quad \chi = \frac{g^2}{\Delta}. \end{aligned} \quad (22)$$

Energy eigenvalue errors start at fourth order. With ground-sector calibration $\omega_g = \omega - \chi > 0$, the excited-sector rate is

$$r_{e/g}^{(2)} = \frac{\omega + \chi}{\omega - \chi} = 1 + \kappa_{\text{eff}}, \quad \kappa_{\text{eff}} = \frac{2g^2/\Delta}{\omega - g^2/\Delta}. \quad (23)$$

Positive and negative detuning yield opposite signs. The interaction does not force denser matter to slow the clock.

The exact fixed-excitation block in $\{|N, g\rangle, |N-1, e\rangle\}$ is

$$H_N = \begin{pmatrix} N\omega & g\sqrt{N} \\ g\sqrt{N} & (N-1)\omega + \Omega \end{pmatrix}, \quad E_{N,\pm} = N\omega + \frac{\Delta}{2} \pm \frac{1}{2}\sqrt{\Delta^2 + 4g^2N}. \quad (24)$$

The branches connected to the bare states are

$$E_g(n) = n\omega + \frac{\Delta}{2} - \frac{\text{sgn } \Delta}{2}\sqrt{\Delta^2 + 4g^2n}, \quad (25)$$

$$E_e(n) = (n+1)\omega + \frac{\Delta}{2} + \frac{\text{sgn } \Delta}{2}\sqrt{\Delta^2 + 4g^2(n+1)}. \quad (26)$$

The first formula also gives $E_g(0) = 0$. From $0 \leq 1 + u/2 - \sqrt{1+u} \leq u^2/8$ for $u \geq 0$,

$$|E_{\text{exact}} - E_{\text{second order}}| \leq g^4 N^2 / |\Delta|^3. \quad (27)$$

For $\omega = 1, g = .03, \Delta = \pm .5$, exact $N = 1, 2, 3$ blocks verify this bound and both rate directions: the second-order ratios are 1.00360649 and 0.99640647. These do not realize $\kappa = 1/4$.

Beyond second order the clock gaps depend on n ; the spectrum is not an exact rigid rescaling. Bare N_D is not conserved by H_{JC} : the effective conserved sectors are dressed, with correspondingly transformed density and readout operators. Thus the microscopic check supports the possible sign and dispersive structure, not an exact identification of the bare two-level density with a universal clock rate.

6 Closure, physical interpretation and the next useful decision

The requested sequence is complete for one declared candidate:

1. A positive, ground-subtracted density operator and its reference measure are specified without assuming a spatial volume.
2. One interaction is given. A common stationary constraint and matched finite state yield normalized joint clock probabilities.
3. Relative rate follows from those probabilities under fixed calibration; common rescaling cannot create a differential effect.

What has been obtained is an *environment-dependent clock-frequency shift*. Calling it universal slowing of time would require additional physics. In particular, the coupling's sign and dependence on the clock transition are not fixed by the density definition. Source fluctuations can reduce visibility rather than define one deterministic time map.

What remains an assumption, not a hidden theorem

Ordinary quantum mechanics, a subsystem split, the operational cell, the phase POVMs, reference shielding, preparation and commensurate spectra are inputs. No energy-information identity, negative energy, mass conversion, emergent space, metric or universal proper time follows. The finite reference cannot supply an exact canonical time operator. Conditional phase arcs or record/epoch labels remain necessary at recurrences, as established by G-R2.

The G-R2 finite-history recording gates and the present energy constraint are *not yet one autonomous energy-complete recording model*. No detector design or physical memory lifetime is claimed. The exact stationary test does not silently extend to arbitrary reference interactions, irrational spectra, or an actual multilevel transmon.

Two physically useful follow-ups, not further fits

First, select a source and clock realization that fixes the interaction and its sign independently of a desired slowdown. If an intensive density rather than total excitation energy is essential, compare differently sized sources at fixed density with the same specified coupling profile. Second, test whether independent clock transitions acquire the same fractional shift. Transition-dependent distortions support ordinary clock perturbation; a universal time interpretation needs a principle explaining why these distortions disappear across different clocks. Neither task is replaced by fitting a free function of E_d .

Executable verification

The companion `verify_gr3_density_clocks.py` passes 273 checks: stationarity, positivity, conditional states, both POVMs, invariant effects, two-clock likelihoods, relative slopes, energy-unit and zero-shift invariance, the failed mean-density substitution, and exact versus dispersive exchange spectra. Previous files are hashed. All 4636 Route-G regression checks pass. These are algebra/implementation checks, not experimental confirmation.

References

- [1] A. R. H. Smith and M. Ahmadi, *Quantizing time: Interacting clocks and systems*, Quantum **3**, 160 (2019), arXiv:1712.00081v3.
- [2] A. Blais, A. L. Grimsmo, S. M. Girvin and A. Wallraff, *Circuit quantum electrodynamics*, Rev. Mod. Phys. **93**, 025005 (2021), Sec. III.B and Appendix B.2.b, doi:10.1103/RevModPhys.93.025005.